Task 1. Self-driving cars like Tesla Autopilot

Yes, self-driving cars like Tesla Autopilot **use** Artificial Intelligence (AI), as it mimics the human driving capabilities.
The AI senses the environment, makes decisions and controls the vehicle without any human intervention.

- Self-driving cars use neural network which is a subfield of AI.
- It gets the input from car's sensors and cameras
- It recognizes road signs, traffic lights, pedestrians, and other vehicles
- It predicts the behaviour of nearby objects and plans the driving paths.
- Thus, it makes real-time decisions.

Perception and decision-making is driven by Artificial Intelligence.

Self-driving cars make extensive use of **Machine Learning** (ML). All three types, 'Supervised Learning', 'Unsupervised Learning' and 'Reinforcement Learning' are used in case of self-driving cars.

1) **Perception: Learn to Recognize the Environment**
   Cameras are used to detect lane markings, traffic lights, pedestrians, and signs.
   Lidar (laser scanning) is used to create a 3D map of the surroundings.
   Radar is used to detect the speed and distance of other vehicles.
   Ultrasonic sensors are used for close-range detection.
   Machine Learning models are trained to process data from all these sensors.
2) **Understanding Situations:** The Machine Learning models help the car understand what is happening. i.e. predicting the behavior using past data. Thus, the car understands the situations like 'Pedestrian crossing the road' etc.
3) **Decision-Making and Planning:** The car uses Machine Learning to decide the next course of action. **Reinforcement learning** is often used which trains the car by giving it rewards for safe, smart behavior in a simulated environment.
4) **Training with Data:** The self-driving cars are trained using millions of miles of driving data, tricky situations, annotated datasets etc.
5) **Continuous Learning:** The self-driving cars keep learning, retraining and improving.

Here,

a) **Supervised Learning** is used for perception. Self-driving cars learn from labeled data like images and use them for object detection, traffic light recognition etc.
b) **Unsupervised Learning** is used for understanding patterns in data. E.g. clustering driving behaviors, anomaly detection, map building.
c) **Reinforcement Learning** is used for decision making and planning. Self-driving cars learn by trial and error, get rewards for good actions and penalties for bad actions. This is used for decision making, path planning etc.

Task 2. Mobile banking app displaying your current account balance

Mobile banking app displaying your current account balance **does not need to use** Artificial Intelligence or Machine Learning.
In order to display the account balance involves retrieving the data from database and displaying it.

- The app makes a query and retrieves account balance from a centralized banking database or server.
- It then displays this value to the user via a graphical interface (UI).
- It's a simple data retrieval task.
- Learning, pattern recognition, prediction, or decision-making is not required to be involved in this core function of displaying current account balance.
- The process follows fixed rules with clear, correct outcomes.
- Using AI here would be inefficient and unnecessary.

Artificial Intelligence is used for the tasks which involve:

- Pattern recognition
- Predictions
- Natural language processing
- Complex decision-making

None of these features are required while querying and displaying account balance of a person. And hence Artificial Intelligence and Machine Learning are not required to be used for the same.

Task 3. Netflix recommending movies and TV shows based on your viewing history

Netflix recommending movies and TV shows based on user's viewing history **use** Artificial Intelligence and Machine Learning capabilities.

Artificial Intelligence analyzes vast amounts of data for each user, such as,

- Viewing history like what user watched, how long he/ she stayed on that, whether user finished it etc.
- Search queries
- Genres and categories
- Ratings or likes/dislikes
- Device types used by the user and times of viewing
- Behavior of similar users

Based on the above data, Netflix builds a detailed user profile. It processes the viewing history, learns from user's habits and predicts what he/she will like. Machine Learning algorithms recommend systems to personalize content for the user that aligns with the user's preferences. Thus, the recommendations are generated.

Netflix uses the combination of mainly **Supervised and Unsupervised Learning** types of Machine Learning:

1) **Supervised Learning:** It is used when the system is trained on labeled data.
    a. Machine learning tracks the user's preferences e.g. user's interactions labelled as 'liked' or 'disliked' etc. This is used for predictions.
    b. Algorithms like logistic regression, neural networks, or decision trees might be used to predict user preferences.
2) **Unsupervised Learning:** It is used for clustering users or content into similar groups.
    a. Machine learning is used to cluster the people who watch similar types of content.
    b. It helps in recommending content based on similarity.
    c. K-means clustering, matrix factorization etc. are examples of the technique.
3) **Reinforcement Learning (used very less) :** It is used in some cases. E.g. System gives some recommendations. If user clicks and watches the movie/ show recommended, it is a 'reward'.

a. Netflix experiments to optimize recommendations over time based on user feedback.
b. The system learns through trial and error, getting a "reward" when users watch or engage with a recommendation.
c. Example, the home page layout is reordered to maximize clicks.

Task 4. Barcode scanner at a supermarket checkout

A barcode scanner at the supermarket checkout **does not need to use** Artificial Intelligence and Machine Learning capabilities. Barcode scanner functionality uses fixed rules and databases without learning or adaptation. It is designed to be a simple, structured encoding system that can be read and interpreted using fixed rules.

A barcode like UPC or EAN is a standardized pattern of lines that represents numbers or characters. The rules for interpreting these patterns are already defined and universal. Thus, scanners do not need to learn how to read them. There is no uncertainty.

The functionality of a barcode scanner is:

- Scanner uses a laser or camera to scan the barcode.
- The scanned barcode is then matched to an item in the store's database.
- The scanner sends the decoded data, generally a product number, to the Point of Sale (POS) system to retrieve the product price and product information.
- This process is entirely rule-based, relying on decoding of barcode patterns and database lookup for retrieving the basic information.
- Decision-making, pattern recognition, or learning is not involved in the above process. Hence Artificial Intelligence is not used.

Task 5. YouTube's content moderation system that flags potentially harmful videos

YouTube's content moderation system that flags potentially harmful videos **uses** Artificial Intelligence and Machine Learning capabilities.

Artificial Intelligence classifiers help detect potentially violative contents at scale and reviewers work to confirm whether content has actually crossed policy lines. AI classifiers are trained on large amounts of data, it uses frameworks such as TensorFlow, PyTorch etc. to analyze video contents, transcripts, comments etc.

Machine learning identifies potentially violative content at scale and nominates for review content that may be against our Community Guidelines. Content moderators then help confirm or deny whether the content should be removed.

The functionality is as below:

- The Machine Learning systems scan uploaded video content, transcripts, comments etc. to detect possible violations of the Community Guidelines.
- YouTube uses Machine Learning "classifiers" trained on examples of violative content to identify new content that looks like it might violate rules. These classifiers check video-metadata like title, description, tags etc., comments, and video/audio/visual features to detect risk.
- The system flags the content for human review. Human moderators check whether the policy is being violated.
- The system prioritizes which content is risky and scales the moderation effort to the huge volume of uploads.
- The system is updated by using the outcomes of human review to improve the ML models.
- The flagging of contents is done early before many people view it.
- Sometimes, automatic removing is done in case of clear violation.

'Supervised Learning' type of Machine Learning is used in YouTube's content moderation system.

- The flagged contents are reviewed by human reviewers, and their decisions are used to train machines for better coverage in future.
- Examples of human decisions like 'Child Safety', 'Hate Speech', 'Spam' etc. are used as labels. These labelled examples are fed to models to learn to classify violative and non-violative contents.
- There is evidence of removing many videos flagged by Machine Learning before any human flag.

YouTube deals with hundreds of hours of video uploaded every minute. With such a huge amount of data, human review alone would be impossible. AI and Machine Learning enable **scalability** by automatically pre-flagging and reduces human burden. The increases **speed** by earlier invention. They also enable **pattern detection** by spotting similarity to previously removed contents, detecting new threat patterns and identifying metadata or visual cues of risk. Thus, AI and Machine Learning is useful in this scenario.